

Mohammed Warish

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PROFILE SUMMARY

AI Engineer with **2+ years** of experience building production AI systems across Agentic AI, Computer Vision, Voice AI, and Generative AI. Experienced in developing end-to-end AI solutions using LLMs, LangGraph, LiveKit, YOLO, DeepStream, Kafka, and AWS across healthcare, smart city, industrial automation, and wildlife domains.

PROFESSIONAL EXPERIENCE

AI Engineer

04-2025 – Present | Hyderabad

Quanteon Solutions

- Developed production-grade **Voice AI agents** for Prior Authorization and Appointment Scheduling using a modular multi-agent architecture with **LiveKit**, integrating real-time **STT–LLM–TTS** pipelines, SIP telephony (IVR/DTMF), interruption handling, and tool-calling workflows powered by **Qwen-3, Gemma, and Amazon Bedrock** models.
- Engineered an end-to-end **LLM evaluation and observability framework** using **Langfuse** and **LLM-as-a-Judge** to evaluate latency, hallucination, task completion, response quality, and conversation accuracy, enabling continuous monitoring and optimization of production AI systems.
- Fine-tuned **Qwen-3** using **LoRA** for healthcare-specific structured output generation and tool-calling, improving the reliability and consistency of production Voice AI agents.
- Developed an **AI-powered Smart City Operational Intelligence Platform** using **DeepStream, YOLO11, ByteTrack, Kafka, and Apache Flink** for real-time CCTV video analytics, event processing, and incident management.
- Implemented computer vision pipelines for **Wrong-Way Driving, Footpath Driving, Road Blockage, Detection, Stray Cattle, Loitering, Intrusion Detection**, along with fine-tuning models for smart city-specific scenarios.
- Developed an **LLM-powered Operational Advisory Agent** using **Amazon Bedrock** to transform computer vision events into contextual, explainable operational recommendations for emergency response and city operations.
- Developed an AI-driven **City Mobility Recommendation System** leveraging **Amazon Bedrock**, predictive analytics, optimization, explainability, and scenario simulation to recommend work-from-office shift schedules for companies while reducing city corridor congestion.
- Developed a **Women Safety Patrolling Recommendation Agent** using historical incidents, hotspot intelligence, and contextual risk factors to generate dynamic police patrol schedules and proactive public safety recommendations.
- Implemented predictive AI workflows for **Ghost Jam Prediction** and **Hyderabad Metro Crowd Surge Forecasting**, enabling proactive traffic management, crowd surge prevention, and operational planning through AI-driven recommendations.

AI ML Engineer

04-2024 – 04-2025 | Hyderabad

Eizen.AI

- Developed a **LangGraph-based multi-agent Voice AI system** for **SRE workflows** — agents capable of natural language querying over **correlation and causation graphs, real-time anomaly summarization, root cause narration, and suggested remediation steps** reducing manual triage effort and enabling **early incident resolution**.
- Architected the ML approach for animal and human **re-identification across 60+ multi-camera environments at Vantara Zoo** — designed system using **Siamese Networks, CNNs, and Swin Transformers**.
- Designed **unsupervised learning pipeline (SimCLR + DBSCAN)** for **elephant vocalization behavior analysis**.
- Built **leopard detection and movement monitoring** system using **YOLOv8** by providing **region-wise behavior analytics**, and real-time monitoring approach for **wildlife safety at Vantara Zoo**.
- Built production **QA system using YOLO** for **real-time object detection in chip manufacturing**, integrated with **Kafka and Jenkins** for automated process control on **anomalous events**.

SDE Intern

11-2023 – 04-2024

Dilate Labs Pvt. Ltd.

- Developed a **customer and lab partner application** using **Node.js, TypeScript, MongoDB, and REST**, integrating **Firestore reCAPTCHA and Google login** for improved security, enhancing user experience **by 10%**.
- Developed **over 40 REST APIs** for functionalities including **Google Maps APIs, Secure Razorpay payment gateway, Order Booking, Wallet Management, Patient Report Card, Cloudinary File upload and Reviews**.

EDUCATION

BTech in Computer Science and Engineering

2020 – 2024 | Chittoor Andhra Pradesh

IIIT Sri City Chittoor Andhra Pradesh

TECHNICAL SKILLS

Programming Languages : Python | JavaScript

Libraries & Frameworks : PyTorch | TensorFlow | Scikit-learn | Pandas | NumPy | Fast API | Node Js

AI & LLM : Agentic AI | Multi-Agent Systems | RAG | Prompt Engineering | Prompt Chaining | LoRA Fine-tuning | PEFT | Structured Output | Tool Calling | Embeddings | Claude | Amazon Bedrock | Qwen-3 | Gemma | Mistral | Phi-3 | vLLM | Ollama
Agent Frameworks : LangGraph | LangChain | AutoGen | Livekit
Computer Vision : YOLO11 | DeepStream | OpenCV | OCR | Swin Transformer | Vision Transformer (ViT) | ResNet | VGG16
Data & Databases : Kafka | Flink | PostgreSQL | Redis | MongoDB | MySQL | FAISS
Cloud & DevOps : AWS (Bedrock, EC2, S3) | Azure OpenAI | Docker | Jenkins | Git
Voice AI : Cartesia | Deepgram | Whisper | ElevenLabs | Kokoro | Azure Speech | SIP Telephony | IVR (DTMF) | Speaker Diarization | Silero VAD | Krisp Noise Cancellation
Observability & Evaluation : Langfuse | LangSmith | LLM-as-a-Judge

PROJECTS

Interviewer Agent

09-2025 – 11-2025

AI-driven Real-Time Interviewing & Candidate Evaluation System

- **Tech Stack:** LiveKit | Deepgram | Elevenlabs | Azure OpenAI | AWS bedrock | PostgreSQL | Redis
- Built an **autonomous AI interviewer agent** for real-time candidate interaction, enabling dynamic question generation, **contextual follow-ups**, and natural turn-taking with interruption handling.
- Developed a **stateful conversation engine** to manage interview flow, track candidate responses, and **adapt questioning strategy dynamically**.
- Implemented **end-to-end session orchestration and LLM as judge evaluation pipeline**, generating structured candidate reports including **performance insights and behavioral metrics**.
- Built scalable real-time state management utilising **PostgreSQL** for **session persistence** and **Redis** for **caching**.
- **Reduced manual screening effort by ~80%** through automated interview conduction and AI-driven evaluation.

Multimodal AI Therapist Agent

02-2025 – 04-2025

AI-driven Therapeutic Conversation System for Trauma Support

- **Tech Stack:** LLaMA2 | Phi-3 | LangGraph | (Amazon Polly) TTS | Computer Vision | SQL
- Designed and developed a **multimodal AI therapist agent** to **assist trauma survivors** (e.g., trafficking victims) by delivering structured, empathetic therapy aligned with **TRT/CBT methodologies**.
- Built a **stateful, multi-agent orchestration pipeline using LangGraph**, enabling dynamic control flow across components such as emotion detection, state tracking, and response generation.
- Integrated a **multimodal emotion detection system** combining **facial expression analysis and acoustic speech emotion recognition**, with real-time signals streamed via Kafka and injected into agent context.
- Developed a **therapy policy engine** to constrain LLM outputs, ensuring responses adhere to therapeutic protocols.
- Achieved **low latency real-time voice interaction** using whisper and Amazon Polly for natural conversational experience.
- Designed persistent storage using **SQL** to track session states, therapy progress, and conversation history across interactions.

Multi-Agent AI Project Management System

08-2024 – 09-2024

Revolutionizing Project Management with AI Agents and LangGraph

- **Tech Stack:** LangChain | LangGraph | FastAPI | RAG | GROQ API | FAISS | SQLAlchemy | Docker
- **Automated task generation and prioritization** using **AI-driven Task and Priority Agents** based on project description.
- Implemented **RAG with FAISS** for similarity search and language models **langchain_ollama, langchain_groq** to provide **context-aware task suggestions, reports, and recommendations**.
- **Suggested optimal team configurations and strategies** by analyzing team capabilities and project requirements.

CERTIFICATIONS

GPS Spoof and Detect in Ardupilot Simulating UAVs

01-2023 – 08-2023

- **Published** a paper titled "GPS Spoof and Detect in Ardupilot Simulating UAVs" at the **21st IEEE OITS International Conference on Information Technology**.
- Demonstrated **problem-solving** skills by successfully **solving over 200 DSA problems on Leetcode**.
- **Generative AI Fundamentals** – Databricks Academy
- **AI Agent Fundamentals** – Databricks Academy
- **Advance MCP Servers for Automation in Production** - Huggingface